Data Analysis Report

shopping_mall_dirty_data.csv

95.7/100

Data Quality Score

Improved from 94.6 → 95.7

Executive Summary

	15	7	100.0%
Clean Rows	Columns	Numeric Fields	Completeness

The dataset spans 7 numeric, 7 categorical, and 1 date field(s). Post-cleaning completeness is 100.0%. Data quality score improved from 94.6 → 95.7 / 100.

Cleaning Actions Performed

• Standardized 12 column name(s) to snake_case

• Stripped currency symbols from 2 column(s): price, total_sales

• Normalized text casing and whitespace in 6 column(s)

• Imputed 60 missing values across 2 column(s) using smart per-column strategies

• Removed 20 exact duplicate row(s)

• Parsed 1 date column(s) and created 3 derived time features

• Capped 24 outlier(s) across 2 column(s) using IQR Tukey fences

Data Overview

Field Name	Type	Role	Non-Null	Unique	Sample Value
order_id	object	id	520	520	ORD0230
customer_name	object	text	520	485	Kayla Levine
city	object	text	520	488	Lake Teresa
product_type	object	categorical	520	14	Hoodie
brand	object	categorical	520	11	H&M
quantity	int64	numeric	520	15	3
price	float64	currency	520	511	342.96
total_sales	float64	currency	520	507	1028.88
rating	float64	numeric	520	41	2.5
month	object	categorical	520	12	December
payment_method	object	categorical	520	4	Net Banking
order_date	datetime64[n	date	520	284	2025-12-04 00:00:00
order_year	int32	—	520	2	2025
order_month	int32	—	520	12	12
order_quarter	int32	—	520	4	4

Descriptive Statistics

Statistic	quantity	price	total_sales	rating	order_year	order_month	order_quarter
mean	7.95	2,359.21	18,551.83	2.94	2,025.00	6.82	2.60
std	4.34	1,552.00	17,552.16	1.12	0.04	3.43	1.13
min	1.00	-2,413.47	-27,618.14	1.00	2,025.00	1.00	1.00
25%	4.00	1,211.90	6,177.18	2.00	2,025.00	4.00	2.00
50%	8.00	2,350.76	14,800.69	2.90	2,025.00	7.00	3.00
75%	12.00	3,628.80	28,707.39	3.90	2,025.00	10.00	4.00
max	15.00	4,975.34	62,502.71	5.00	2,026.00	12.00	4.00

Data Quality — Before vs After

Metric	Before Cleaning	After Cleaning	Improvement
Total Rows	540	520	**–20**

Missing Values	60	0	–60
Duplicate Rows	20	0	–20
Quality Score	94.6/100	95.7/100	+1.1 pts

Missing Value Imputation Detail

Field	Count	% Missing	Strategy	Fill Value
customer_name	35	6.5%	mode	Amy Tyler
city	25	4.6%	mode	New Stephanie

Key Findings

Data Quality

■ Quality score improved by 1.1 points (94.6 → 95.7/100)

The cleaning pipeline significantly improved dataset reliability. Key contributors: missing value imputation, duplicate removal, and outlier capping.

■ 60 missing values filled across 2 field(s)

Imputation strategies applied: mode. Fields treated: customer_name, city. Median was preferred over mean for skewed distributions.

■■ 20 duplicate record(s) removed

20 exact duplicates and 0 business-key duplicates were detected and removed to prevent double-counting in aggregations.

■ 24 outlier(s) detected and capped in 2 field(s)

"Total Sales" had the most outliers (15, 2.9% of rows). Values were capped using IQR Tukey fences rather than deleted, preserving sample size while limiting distortion.

■ 2 currency field(s) standardized

Currency symbols and formatting stripped from: price, total_sales. Values are now clean floats ready for aggregation.

■ 3 date-derived feature(s) created

Temporal features added: order_year, order_month, order_quarter. These enable year-over-year, quarterly, and monthly analyses.

Statistical Highlights

■ "Price" shows high variability (CV = 66%)

Range: -2.4K to 5.0K. Coefficient of Variation = 66% indicates this field is highly dispersed — segment analysis may reveal subgroup patterns.

■ "Total Sales" shows high variability (CV = 95%)

Range: -27.6K to 62.5K. Coefficient of Variation = 95% indicates this field is highly dispersed — segment analysis may reveal subgroup patterns.

■ "Order Year" is right-skewed (long right tail)

Skewness = 22.80. Mean (2.0K) differs from median (2.0K), meaning high values are pulling the mean above the median. For modeling: consider log transformation.

Distribution Analysis

■ "Quantity" is not normally distributed (Shapiro p=0.007)

p-value = 0.0067. Median-based statistics and non-parametric tests are more appropriate.

■ "Price" is approximately normally distributed (Shapiro p=0.057)

p-value = 0.0568. This supports using mean-based statistics.

■ "Total Sales" is not normally distributed (Shapiro p=0.001)

p-value = 0.0013. Median-based statistics and non-parametric tests are more appropriate.

Correlation Analysis

■ Strong correlation: "Price" & "Total Sales" (r=0.73)	These fields are positively correlated (|r| = 0.73). 1 additional strong pair(s) found. In ML models, consider removing one to reduce multicollinearity.
■ 1 moderate correlation(s) identified	Pairs: "Quantity" & "Total Sales" (r=0.57). These relationships may be analytically useful.

Field Analysis

■ "Order Id" appears to be a unique identifier	Every value is distinct — this column functions as a primary key. Exclude from groupby, aggregations, and model training.

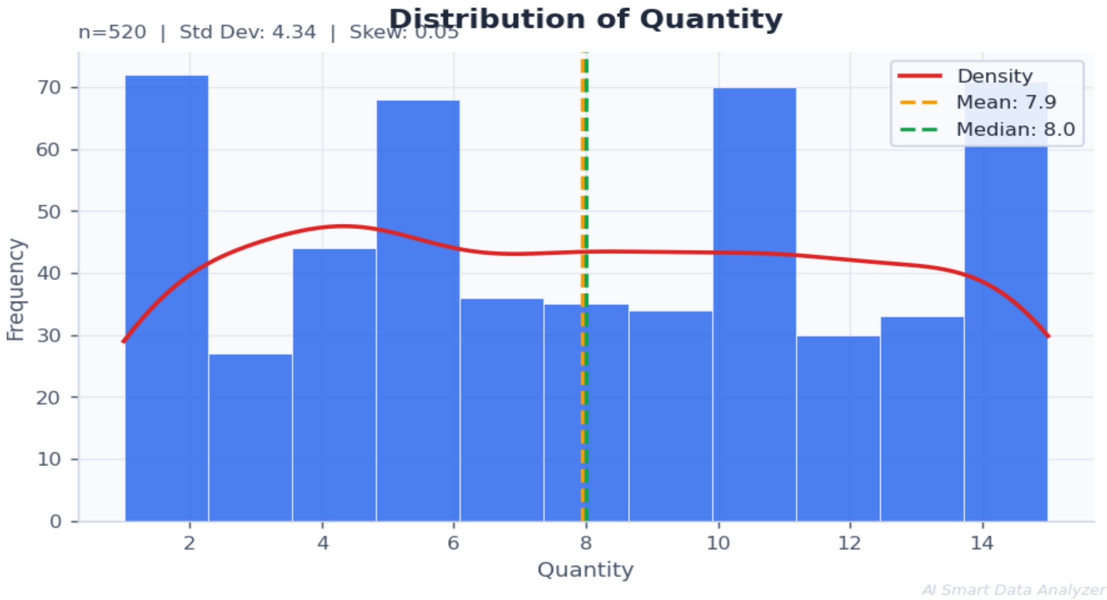
Time Analysis

■ "Order Date" spans 364 days of data	Date range: 02 Jan 2025 to 01 Jan 2026. Time-series analysis, trend forecasting, and seasonal decomposition are applicable.

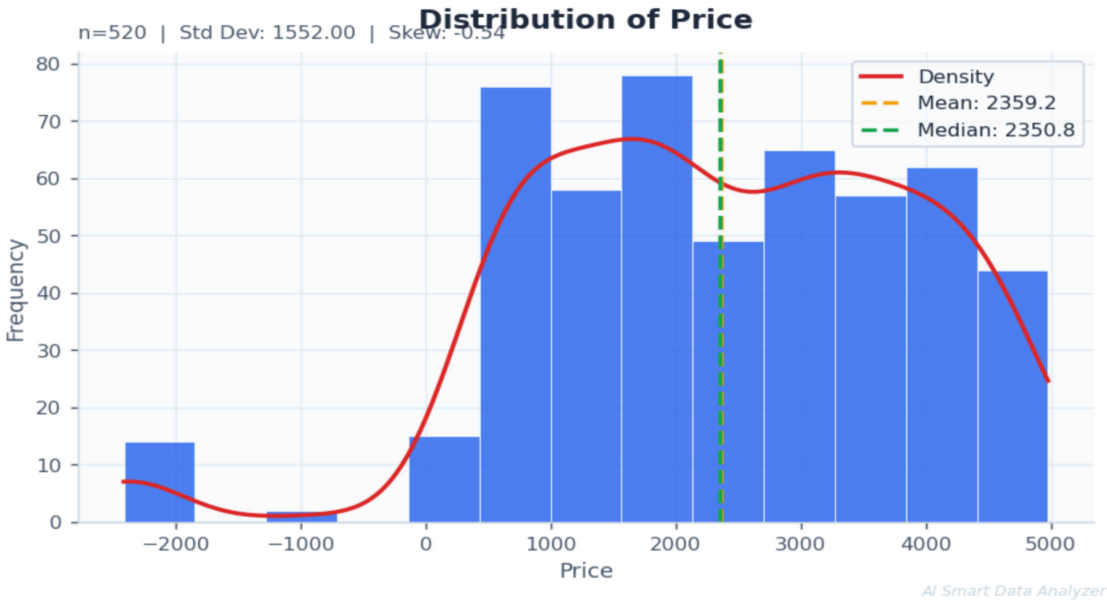
Recommendation

■ Dataset is ready for classification or regression modeling	With 7 numeric and 7 categorical fields, this dataset supports supervised ML. Suggested next steps: encode categoricals, scale numerics, split train/test, and evaluate with cross-validation.
■ Time-series modeling and trend analysis are viable	Date fields are present and parsed. Consider ARIMA, Prophet, or simple moving averages for forecasting. Derived date features (year, month, quarter) have been added to support this.

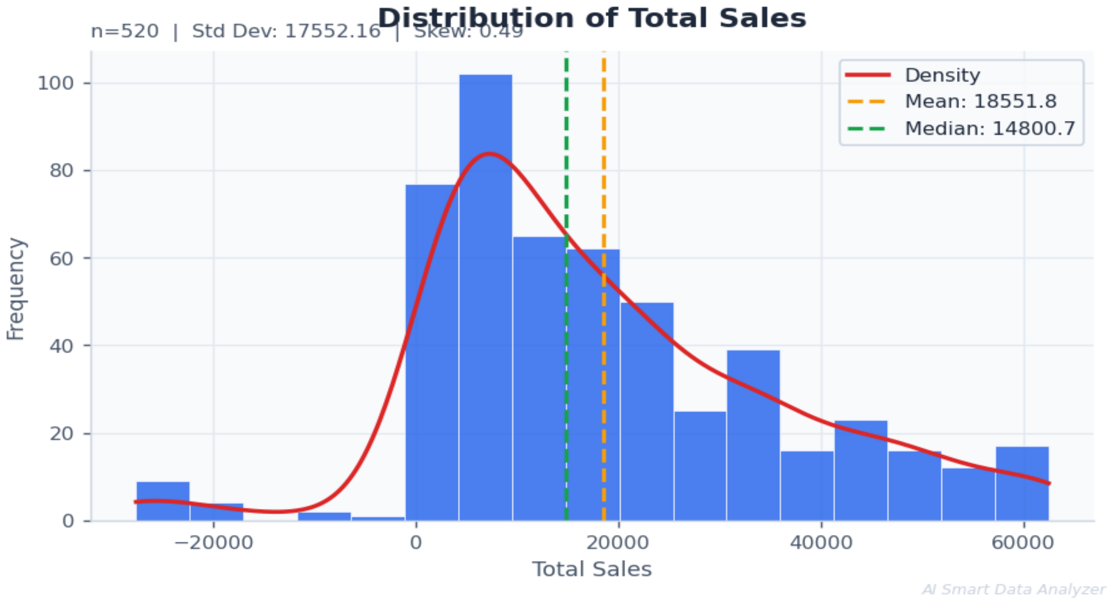
Visualizations



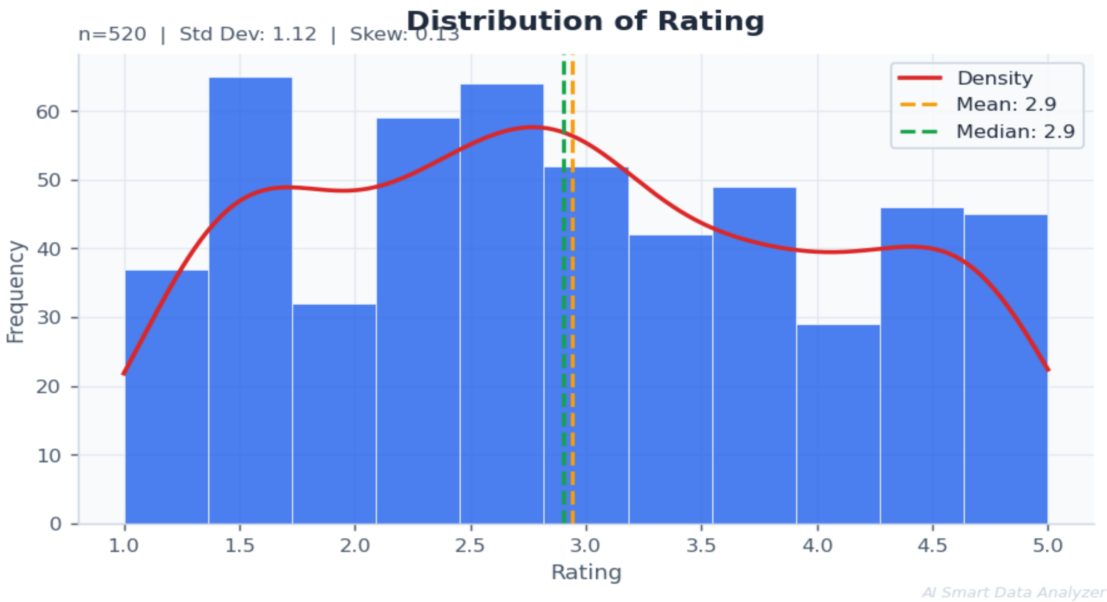
Distribution of Quantity



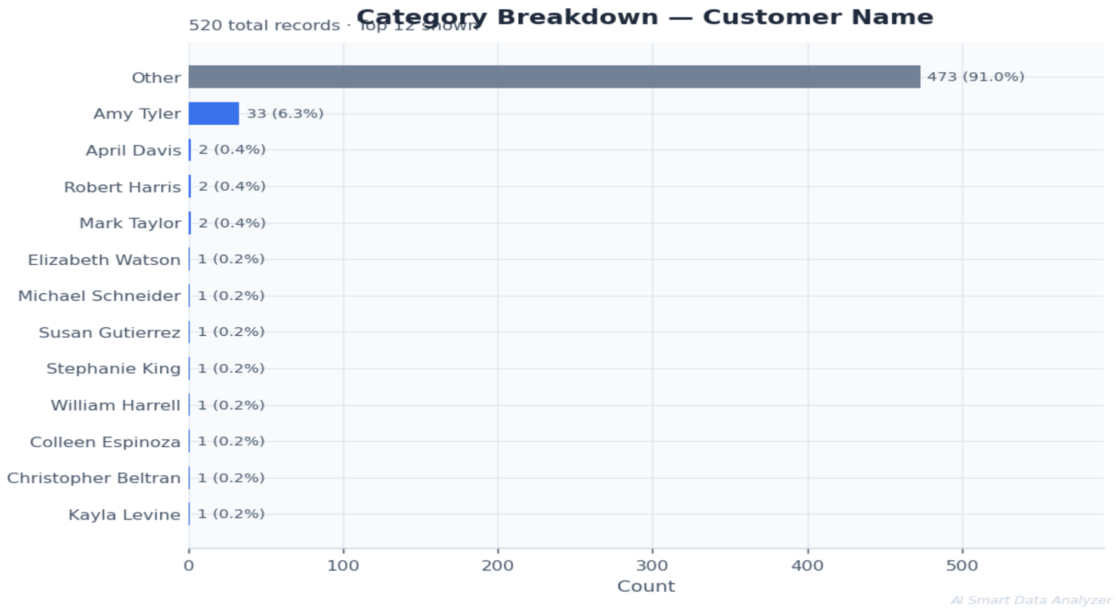
Distribution of Price



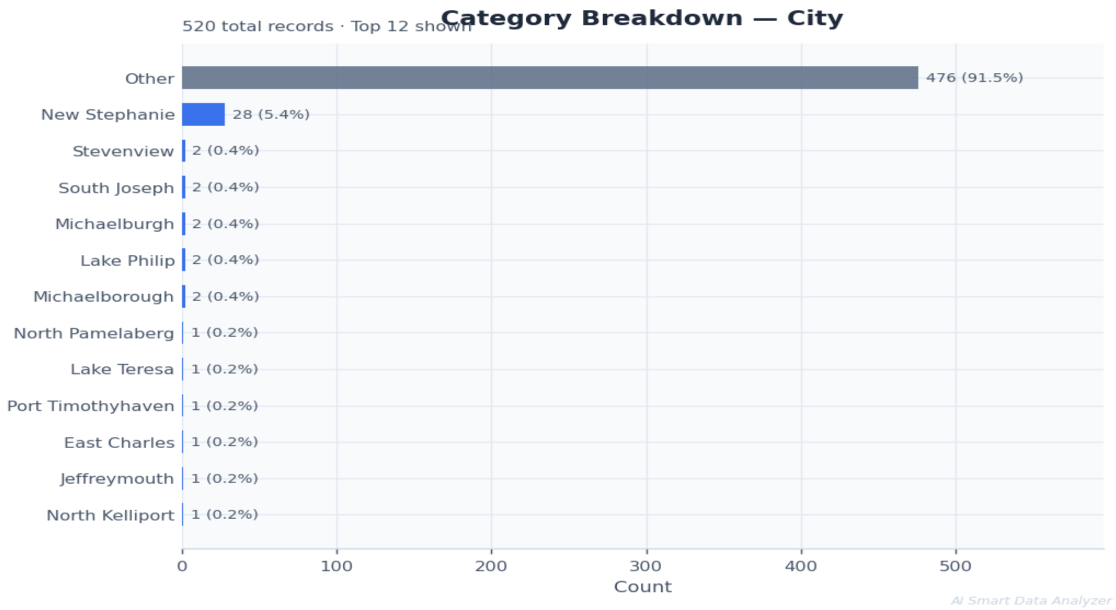
Distribution of Total Sales



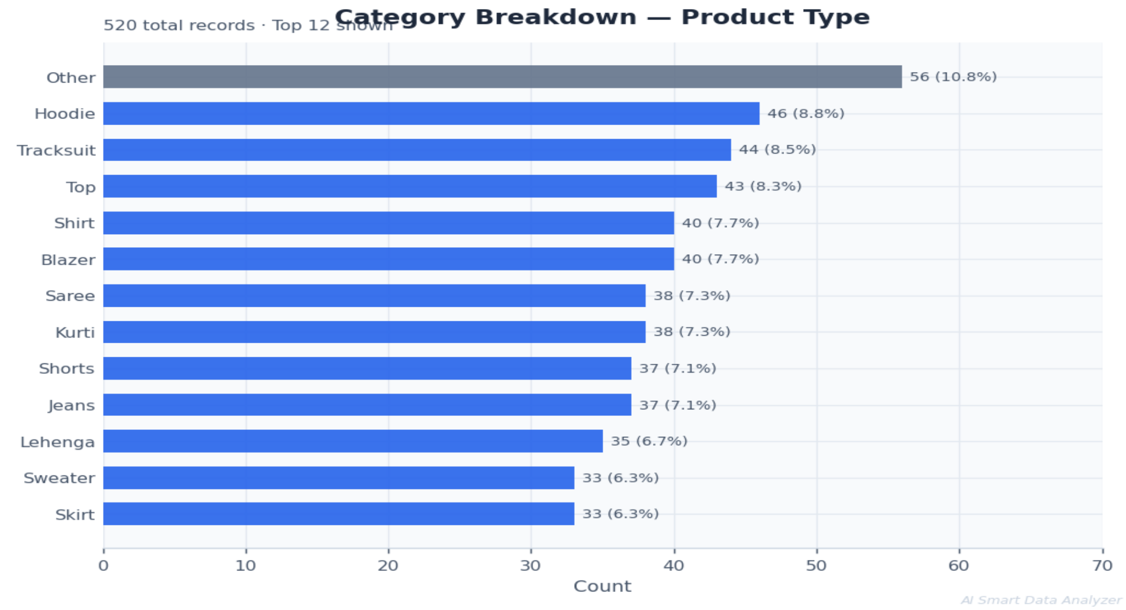
Distribution of Rating



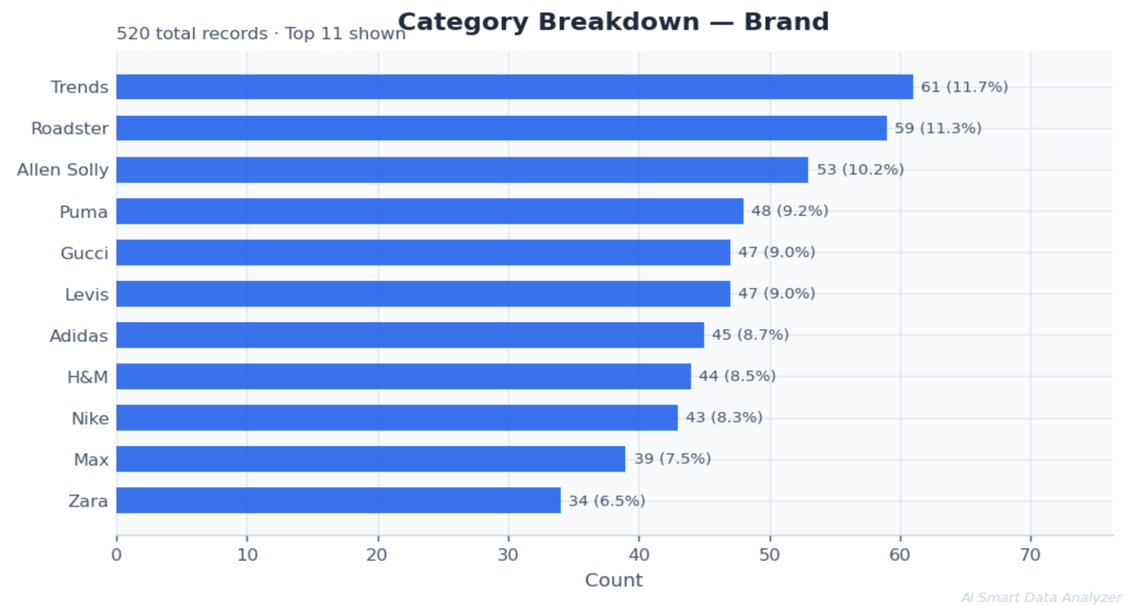
Category Breakdown — Customer Name



Category Breakdown — City



Category Breakdown — Product Type



Category Breakdown — Brand

Methodology & Technical Appendix

This report was generated by the AI Smart Data Analyzer pipeline. The following methodology was applied:

Schema Normalization

Column names standardized to snake_case. Data types classified by content analysis into roles: numeric, categorical, date, currency, percentage, identifier, email, phone.

Missing Value Imputation

Strategy selected per column: median for skewed numeric distributions, mean for near-normal distributions, mode for low-sparsity categoricals, "Unknown" for high-sparsity categoricals, forward/backward fill for temporal fields.

Duplicate Resolution

Exact row duplicates removed. Business-key duplicates identified using ID + date combinations where present.

Text Normalization

Hidden Unicode characters removed. Categorical labels normalized to Title Case. Whitespace standardized. Fake null strings (null, n/a, —, etc.) converted to NaN.

Date Parsing & Enrichment

Mixed date formats parsed via infer_datetime_format. Derived temporal features (year, month, quarter) created for columns with ≥10 unique dates.

Outlier Treatment

IQR Tukey fences applied (Q1 – 1.5×IQR, Q3 + 1.5×IQR). Z-score cross-validation performed. Values capped (Winsorized) rather than deleted. Identifier and protected columns excluded.

Quality Scoring

Composite score out of 100: Completeness (40 pts), Uniqueness (20 pts), Consistency (30 pts), Validity baseline (10 pts).

Column Renames Applied

Original Name	Standardized Name
Order_ID	order_id
Customer_Name	customer_name
City	city
Product_Type	product_type
Brand	brand
Quantity	quantity
Price	price
Total_Sales	total_sales

Rating	rating
Month	month
Payment_Method	payment_method
Order_Date	order_date

Report generated: 09 May 2026 at 19:23:03

AI Smart Data Analyzer — Portfolio Project